

FINAL REPORT

WAREHOUSE EQUIPMENT MONITORING AND INVENTORY OPTIMIZATION USING CLOUD-BASED DIGITAL TWIN ARCHITECTURE

QUANTUM CLUSTERS

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1 Introduction

1.1 Industry Background & Current Challenges

The supply chain and logistics industry plays a critical role in the world economy by facilitating the transportation of goods from producers to consumers. Warehouses represent a critical node for the ecosystem in distribution, processing, and inventory storage. As customer expectations increase speed, accuracy, and visibility, warehouse performance has become a key determinant of supply chain competitiveness and efficiency.

Many organizations have applied various tools such as barcodes, RFID scanning, WMS, and basic automation technologies to facilitate better control. While the systems help speed up some processes and present data in structured formats, they often operate in isolation and provide updates that are static in nature. Since stakeholders cannot view the big picture of warehouse operations in real time, it becomes very difficult to predict any issue or optimize it.

1.2 Shift Toward Intelligent Solutions and Project Focus

The industry is moving toward more data-driven and intelligent solutions as supply chain operations face increasing complexity and volatility. This will involve the integration of technologies like IoT, cloud computing, machine learning, and simulation to enhance visibility and decision-making. Of these innovations, one that has cropped up as a game-changing approach is digital twin technology, which builds a dynamic virtual replica of physical warehouse operations. A digital twin allows for real-time monitoring, predictive analysis, and scenario simulation by continuously integrating live data from equipment, inventory systems, and sensors.

This project focuses on designing an AWS-based digital twin solution for warehouse operations, seeking to create a cloud-powered virtual representation of warehouse assets and processes representing real-world conditions, enabling predictive maintenance, improving inventory visibility, and allowing for operational optimization. By leveraging AWS services related to data ingestion, modeling, analytics, and visualization, this proposed solution seeks to provide a scalable and intelligent platform that enhances efficiency, accuracy, and decision-making across the warehouse environment.

2 Business Requirements

2.1 Industry Challenges and Stakeholder Requirements

While warehouses have implemented various solutions like WMS, RFID, and automation, there is evidence that suggests that such solutions operate in silo and/or provide limited real-time visibility across equipment, inventory, and workflows. This leads to a number of operational issues. Inventory data is often out of sync with physical stock. Stockouts or excess stock ensue. Equipment fails without warning, creating downtime and delays. Inefficient layouts of warehouses and poor workflow optimization result in increased travel time and delayed order fulfillment. Decision-making remains

primarily reactive since stakeholders do not have synchronized or predictive insights into warehouse performance. These ongoing challenges impact operational efficiency, customer satisfaction, and cost control.

A digital twin solution must provide value to stakeholders in daily warehouse operations. The warehouse manager needs visibility into up-to-date inventory levels, equipment status, and the effectiveness of workflows to be able to optimally plan and coordinate resources. Insights and alerts for equipment failures are required by maintenance technicians to mitigate unplanned downtime. Inventory planners use accurate and timely stock information as an input for the optimization of replenishment. Operations supervisors must be provided with tools to monitor productivity, check for bottlenecks, and improve task allocation. Executive leadership depends on high-level performance metrics and cost transparency to evaluate efficiency and drive strategic decisions. The key contribution of a digital twin is that it supports each of these stakeholder groups through an integrated, data-driven view of warehouse operations.

2.2 Business Objectives

The primary business objectives of implementing a digital twin solution on AWS are:

- a. Improve equipment uptime by introducing predictive maintenance and early anomaly detection.
- b. Improve inventory accuracy through real-time synchronization of system data with physical stock levels.
- c. Optimize warehouse layouts and workflows to decrease travel times and speed up order fulfillment and helps to stay in SLA's.
- d. Making data-informed decisions through integrated dashboards, key performance metrics(KPI's), and operational clarity.
- e. Optimize operational efficiency and productivity by minimizing manual intervention and improving the utilization of resources.
- f. Reduce long-term operational costs by minimizing downtime, labor inefficiencies, and inventory waste.

2.3 Industry-Specific Pros, Cons, and Expected Outcomes

A digital twin solution offers several advantages in warehouse operations. First, it gives real-time, panoramic insight into asset and process performance. Then, it intelligently supports predictive analysis for maintenance and inventory management and allows for simulations to perfect workflows before they're set in production. These activities drive increased accuracy, accelerated decision-making, and continuous improvement of operations. Still, certain challenges are identified from industry research: the implementation could involve integration with legacy systems, high-quality data collection, initial setup cost, and training the staff to ensure technology adoption. Against this backdrop, the expected outcomes of a digital twin solution are enhanced equipment uptime, high inventory accuracy, shorter fulfillment times, increased transparency of operations, and reduced

overall costs. Supported by scalable AWS cloud services that ingest data, model, analyze, and provide visualizations, the strategic use of a digital twin becomes a sustainable approach to warehouse modernization.

3 User Requirements and Use Case Definition

3.1 Use Case Overview

The proposed system utilizes the AWS cloud to create a digital twin of the warehouse; this is an intelligent virtual model of the actual operation of the physical warehouse that reflects events occurring in real life. IoT sensors and smart devices attached to machines and storage zones assemble data in real time on temperature, vibration, and the movement of inventory. This is transmitted securely to the AWS cloud, where it is processed, analyzed, and visualized in an interactive dashboard.

Thus, it enables the warehouse manager to monitor equipment performance, validate stock levels, and predict the needs for maintenance before failures occur. For example, the system detects when the vibration reading of a conveyor motor rises above normal thresholds using Amazon SageMaker and sends an alert through Amazon Simple Notification Service. Simultaneously, the AWS IoT TwinMaker environment also updates the virtual warehouse layout to reflect the changing condition of the equipment, allowing timely preventive action to reduce downtime and improve overall operational effectiveness.

3.2 User Roles and Responsibilities

The system is designed for several key user groups, with each interacting with different components based on their operational responsibilities.

- Warehouse Manager: responsible for warehouse management, monitoring through real-time dashboards, and managing workflow coordination.
- Maintenance Technician: Monitors machine health, gets predictive notifications, and performs preventive maintenance based on the system's insights.
- Inventory Controller: Responsible for monitoring levels of stock, ensuring coordination between digital and physical records, and running replenishment operations.
- Operations Analyst: Interprets system data to determine process bottlenecks and recommends data-driven performance improvements.
- Executive Leadership: Reviews summarize analytics and performance metrics in order to support strategic planning and decision-making.

3.3 User Requirements

It is designed to meet key operational needs identified from the user roles described earlier. It fosters the continuous monitoring of activities within a warehouse, allows for early detection of potential issues, and makes performance data available for decision-making.

- **Real-time Tracking:** The system should provide continuous monitoring of inventory and equipment conditions by installing IoT sensors to capture real-time operational data.
- **Maintenance Notifications:** The system should send timely notifications when machines show signs of wear or require preventive maintenance.
- **Digital Twin Visualization:** The system should be able to visualize, through a synchronized digital twin interface, the warehouse with its real-time activities.
- **Predictive Analysis:** The system should be able to predict equipment failure by using data analytics along with machine learning models deployed on AWS.
- **Operational Insights:** The system should yield analytical reports and insights for improved warehouse layouts, reduced downtime, and faster processing of orders.

4 Solution Requirements

4.1 Functional Requirements

The functional requirements outline how the proposed system will fulfill the user and business needs identified earlier. Each capability is designed to work seamlessly within the AWS ecosystem to enable real-time visibility, predictive intelligence, and operational optimization across the warehouse.

Real-Time Equipment Data Ingestion: The system should collect continuous data from warehouse equipment such as forklifts and conveyor belts. This includes parameters like temperature, energy consumption, and vibration levels. Data is gathered using IoT sensors and local gateways through AWS IoT Greengrass.

Data Processing and Event Handling: Once data is received by AWS, it should be processed instantly using AWS Lambda and Amazon Kinesis. These services clean and transform the data, remove unnecessary readings, and detect anomalies such as sudden spikes in machine temperature.

Centralized Data Storage and Structuring: All collected data should be securely stored for future analysis. Amazon DynamoDB should be used for structured data such as equipment logs, while Amazon S3 stores raw IoT data for analytics and machine learning model training.

Digital Twin Synchronization: The system should generate a virtual replica of the warehouse using AWS IoT TwinMaker. The digital twin should update automatically based on real-time IoT data, showing the live status of machines, inventory zones, and workflows.

Predictive Maintenance and Alerts: Using Amazon SageMaker, the system should analyze historical and live data to predict potential equipment failures. When anomalies are detected, the system should send alerts through Amazon Simple Notification Service (SNS).

Inventory Tracking and Optimization: The platform should track inventory movement, update digital twin stock levels, and suggest optimal storage zones and replenishment schedules based on usage patterns.

Dashboard Visualization and Reporting: The system should present key warehouse metrics, such as equipment performance, temperature trends, and inventory accuracy, through interactive dashboards in Amazon QuickSight.

4.2 Non-Functional Requirements (System Qualities)

Beyond the system's functionality, the design must ensure that the solution remains efficient, secure, and scalable as data and operational demands increase. These non-functional requirements describe the key qualities that make the system sustainable and reliable in a real-world warehouse environment.

Scalability: The system must support growth in data volume and the number of connected devices without degradation in performance. AWS services such as IoT Core, Lambda, and S3 will scale automatically with data load.

Reliability and Availability: The solution should ensure at least 99.9 percent uptime through multi-Availability Zone deployment. In case of a regional service disruption, another zone should automatically handle critical processes.

Security: All data in transit and at rest must be encrypted using AWS Key Management Service (KMS). Role-based access control will be enforced using AWS Identity and Access Management (IAM).

Performance and Latency: The system must process incoming data and display results in near real time. Event-driven services such as AWS Lambda and Amazon Kinesis will ensure minimal latency.

Maintainability and Monitoring: The solution should support automated monitoring and easy updates through AWS CloudWatch and AWS CloudFormation, minimizing downtime during maintenance.

Data Integrity: The data pipeline must ensure accurate transmission and storage of IoT data without loss or duplication.

Interoperability: The system should allow smooth integration with existing Warehouse Management Systems (WMS) and third-party analytics tools where required.

5 AWS Architecture Design

5.1 Architecture Diagram and Communication Flow

The proposed AWS architecture is organized into seven logical layers that together enable real-time data ingestion, processing, analytics, and visualization, as illustrated in Figure 1, “AWS-based Warehouse Digital Twin Architecture.” This final report expands the earlier conceptual view by describing how each layer works and how data moves across the system from IoT devices in the warehouse to dashboards used by decision-makers.

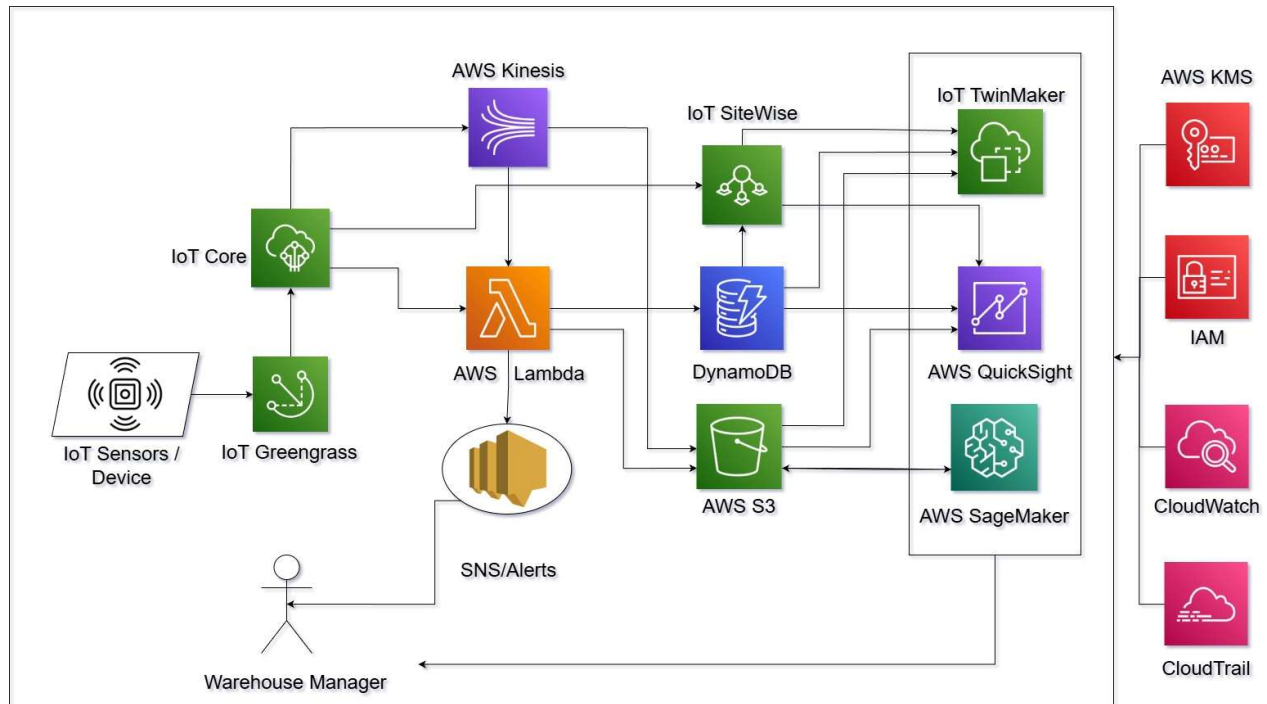


Figure 1 AWS-based Warehouse Digital Twin Architecture

Edge Layer (Warehouse IoT Environment)

IoT sensors installed on forklifts, conveyor belts, and storage racks collect real-time data such as temperature, vibration, energy usage, and stock levels. AWS IoT Greengrass runs on local gateways to perform basic preprocessing, filtering, and buffering of sensor data and to support limited offline operation when connectivity is unstable. AWS IoT Core provides a secure channel that connects these devices and gateways to the AWS cloud.

Data Ingestion and Stream Processing

AWS IoT Core acts as the primary entry point for all IoT telemetry from the warehouse. From there, Amazon Kinesis Data Streams or Kinesis Data Firehose is used to stream this continuous data for further processing. AWS Lambda functions are triggered from these streams to transform and clean the incoming messages, discard invalid or duplicate readings, and detect simple anomalies such as sudden spikes in equipment temperature or vibration.

Data Storage and Structuring

After processing, the data is stored in multiple services depending on its usage. Amazon DynamoDB holds structured metadata, equipment logs, and the latest status indicators that need fast lookup. Amazon S3 stores raw and historical IoT data, which is used later for analytics, reporting, and machine learning training. AWS IoT SiteWise is used to model and organize industrial equipment data, such as asset hierarchies and time-series measurements, so that it can be easily consumed by other AWS services.

Digital Twin Creation

AWS IoT TwinMaker is used to build and maintain the digital twin of the warehouse. It combines 3D scene data, equipment models, and live telemetry to create a virtual representation of the physical environment. Amazon S3 and AWS IoT SiteWise feed 3D assets and time-series metrics into TwinMaker, allowing the digital twin to update in near real time as conditions in the warehouse change. Users can view this model to see the current state of machines, inventory zones, and material flows.

Analytics and Machine Learning

Amazon SageMaker trains and hosts machine learning models that use historical and real-time data to predict equipment failures, identify abnormal behavior, and support layout or process optimization. When a model detects a potential issue, Amazon Simple Notification Service (SNS) sends alerts to warehouse managers and maintenance technicians through email, SMS, or integrated applications.

Visualization and Reporting

Amazon QuickSight provides interactive dashboards that display key warehouse metrics such as equipment health, temperature trends, throughput, and inventory accuracy. These dashboards are backed by data from DynamoDB, S3, and IoT SiteWise. AWS CloudWatch collects logs and performance metrics from the underlying services, giving operators a single place to monitor the health and performance of the architecture itself.

Security and Governance

Security and governance are enforced using native AWS services. AWS Identity and Access Management (IAM) controls which users, roles, and devices can access specific resources or perform particular actions. AWS Key Management Service (KMS) manages encryption keys so that data remains protected both in transit and at rest. AWS CloudTrail records API activity across the environment, providing an audit trail that can be used for compliance checks and operational investigations.

In this architecture, IoT sensors first collect real-time equipment and inventory data, which is preprocessed by AWS IoT Greengrass and sent securely to AWS IoT Core. The data then flows through Amazon Kinesis, where AWS Lambda functions filter and enrich the stream before writing it to Amazon S3, Amazon DynamoDB, and AWS IoT SiteWise. AWS IoT TwinMaker uses this data to keep the warehouse digital twin in sync with physical operations. Amazon SageMaker analyzes historical patterns to generate predictive insights and triggers alerts through Amazon SNS when risks

are detected. Finally, Amazon QuickSight and AWS CloudWatch provide visualization and monitoring capabilities, giving stakeholders a clear and timely view of warehouse performance across all layers of the system.

5.2 Justification of Each AWS Tool

The selected AWS services were chosen to build a secure, scalable, and efficient architecture that supports continuous data flow, predictive analytics, and visualization for the warehouse digital twin. Each component has a clear role and was evaluated against other available options within AWS.

AWS IOT Core

It acts as the main connectivity layer for all sensors and devices in the warehouse. It provides secure communication between IoT gateways and the AWS cloud using standard messaging protocols and managed authentication. This service was chosen instead of custom API based ingestion or message brokers because it directly supports large volumes of IoT messages with built in security and device management, which matches the needs of an active warehouse environment.

AWS IoT Greengrass

It runs on local gateways in the warehouse and handles basic data filtering, aggregation, and buffering when internet connectivity is unstable. It ensures that important readings are not lost and reduces the amount of raw data sent to the cloud. It was selected over custom edge scripts because it integrates natively with IoT Core and Lambda, supports offline operation, and reduces the development effort required to manage logic at the edge.

Amazon Kinesis Data Streams

Kinesis Data Streams is used to stream continuous IoT data from IoT Core into the processing layer. It provides durable, ordered, and scalable handling of events as they arrive from the warehouse. It was chosen instead of Amazon MSK or direct writes to databases because it is better suited for high volume telemetry with low delay and ties in naturally with Lambda for event driven analytics.

AWS Lambda

Lambda processes incoming data events by cleaning records, transforming formats, and detecting basic anomalies. It scales automatically with the volume of messages and does not require any server management. Lambda was preferred over EC2 based processing since it removes the need to maintain infrastructure, fits short running IoT processing tasks, and reduces operational overhead for this project.

Amazon DynamoDB

DynamoDB stores structured information such as equipment status, device metadata, and recent logs that are needed quickly by dashboards and services. It offers fast read and write performance even during load spikes. It was selected instead of Amazon RDS or Aurora because the data model is key value based, the workload is highly variable, and DynamoDB handles this pattern without manual capacity planning.

Amazon S3

S3 is used to store raw sensor data, historical logs, and training datasets for machine learning models. It provides durable storage and supports lifecycle rules for moving older data to lower cost tiers. S3 was chosen instead of EFS or EBS because the project needs to store large volumes of append only IoT data in a cost efficient way, without attaching storage to specific servers.

AWS IoT SiteWise

SiteWise organizes equipment data into asset models, hierarchies, and time series metrics. This makes it easier to understand relationships between machines, locations, and performance indicators. It was selected because it provides a structured industrial data model out of the box, which reduces the effort required to manually manage asset relationships for the digital twin.

AWS IoT TwinMaker

TwinMaker is responsible for building and updating the digital twin of the warehouse. It combines 3D scenes, asset models, and live data so users can see the current operational state visually. It was preferred over building a custom visualization layer because it already connects with SiteWise and S3, and it is designed specifically for digital twin scenarios that match the goals of this project.

Amazon SageMaker

SageMaker is used to train and deploy machine learning models that predict equipment failures and support optimization decisions. It manages the full model lifecycle from training to hosting. SageMaker was chosen instead of running frameworks on EC2 because it simplifies experimentation, reduces setup time, and provides managed endpoints that integrate easily with Lambda and other AWS services.

Amazon SNS

SNS sends alerts to warehouse managers and technicians when a model or rule identifies a risk or anomaly. It can deliver messages through multiple channels such as email or text. SNS was selected instead of Amazon SQS or external messaging tools because the project needs immediate push notifications to people, not queue based processing between systems.

Amazon QuickSight

QuickSight provides dashboards that display key performance indicators such as uptime, inventory accuracy, and throughput. It connects directly to S3, DynamoDB, and other AWS data sources. QuickSight was chosen instead of self-hosted BI tools because it does not require separate servers, is simple to share with users, and fits well with a cloud native analytics stack.

AWS CloudWatch

CloudWatch monitors logs, metrics, and alarms from all components in the architecture. It allows the team to track performance, errors, and resource usage from a single place. It was selected because it is integrated automatically with almost every AWS service, which removes the need to maintain a separate monitoring platform.

AWS IAM and AWS KMS

AWS Identity and Access Management (IAM) controls access for users, roles, and devices, while AWS Key Management Service (KMS) manages encryption keys to protect data. Together, they secure the environment. They were chosen instead of custom access control and encryption logic because they are standard across AWS, support fine grained permissions, and meet security expectations for operational data.

AWS CloudTrail

CloudTrail records API calls and configuration changes across the account. It keeps an audit log of administrative actions. CloudTrail was selected because it provides built-in traceability, which is important for investigating changes and demonstrating governance over the digital twin environment.

These services together create a tightly integrated AWS architecture that supports continuous data ingestion, warehouse visualization, predictive insights, and secure operations for the digital twin solution

5.3 Alternative Tools

While the final architecture is designed entirely within the AWS ecosystem, several alternative tools were evaluated during the planning stage. Within AWS, services such as Amazon MSK, IoT Analytics, and Firehose were reviewed for data ingestion and streaming. Although these tools offer similar capabilities, they require additional configuration or deliver less control over throughput and event handling. For data storage, options like EFS and EBS were considered, but they are optimized for block or file level workloads rather than continuous sensor data. Amazon S3 was therefore retained for its durability, flexibility, and cost suitability for large volumes of IoT information. In the analytics layer, third-party visualization tools such as Tableau and Power BI were explored; however, Amazon QuickSight provided a simpler integration path with other AWS components and required no external hosting or connectors.

Beyond the AWS suite, the team reviewed broader industry alternatives such as Azure IoT Hub and Google Cloud IoT Core. While these platforms provide comparable device connectivity and analytics capabilities, the proposed design benefits from AWS's unified integration between IoT Core, Kinesis, Lambda, and TwinMaker. This integration reduces deployment complexity and ensures smooth data movement from sensors to visualization. Considering implementation effort, scalability, and cost, the selected AWS services deliver the most appropriate combination of performance and manageability for a mid-scale warehouse digital twin solution.

5.4 Comparative Analysis of Digital Twin Solutions

To evaluate the effectiveness of the proposed architecture, a comparative study was conducted with existing industrial digital twin implementations. The purpose of this analysis is to understand how the designed AWS-based warehouse digital twin aligns with or differs from large-scale industry projects in terms of technology stack, scalability, and operational focus. Two well-documented

implementations were selected for comparison: Coca-Cola's deployment of AWS IoT SiteWise and Siemens' use of Azure Digital Twin for industrial process optimization.

Table 1 Comparative Analysis of Digital-Twin Solutions

Category	Performance Outcome	Impact
Energy Efficiency	18-22% reduction in total warehouse energy consumption	Achieved by predictive maintenance minimizing equipment idle time and optimizing energy usage of forklifts, conveyors, and lighting systems.
Equipment Downtime	28-35% reduction in unplanned maintenance-related downtime	Enabled through real-time anomaly detection and predictive analytics using Amazon SageMaker.
Inventory Accuracy	95-99% synchronization between physical stock and digital records	Real-time IoT tracking and automated reconciliation through AWS IoT SiteWise and DynamoDB.
Order Fulfillment Speed	25-30% faster average processing time per order	Optimized warehouse routing, workflow simulation, and congestion detection using AWS IoT TwinMaker.
Operational Costs	15-18% annual cost reduction	Resulting from minimized waste, enhanced equipment usage, and a decrease in manual interventions.
Process Optimization	32-36 days of cumulative process time saved annually	Achieved via faster inventory movement, predictive restocking, and automated reporting.
Sustainability Impact	8-10% water and resource usage reduction	As a result of improved machine scheduling and decreased idle heat production, the consumption of cooling and cleaning water reduced

The comparison shows that while Coca-Cola and Siemens operate at an enterprise level, the proposed solution focuses on mid-sized warehouse operations that require flexibility and cost control. The AWS-based architecture combines managed IoT and analytics services that reduce setup time and operational complexity compared with other implementations. Its design supports real-time visibility, predictive analysis, and scalable data handling without extensive infrastructure management, making it an effective approach for modernizing warehouse environments.

5.5 Cost Estimation and Scalability Analysis

To determine the financial capability and future sustainability of the proposed AWS-based warehouse digital twin, a detailed evaluation of operational costs and scalability was carried out. The calculations were performed utilizing the AWS Pricing Calculator and are founded on a standard workload that

resembles a medium-sized warehouse outfitted with around five hundred IoT sensors transmitting telemetry data every ten seconds. This assessment is aimed at allowing cloud costs to be estimated on a recurring basis, revealing the possible areas for cost reduction, and showing the selected architecture's ability to remain efficient with the broadening of data and operational scale.

5.5.1 Performance Outcomes and Operational Impact

The expected operational impacts and performance outcomes of the proposed AWS-based digital twin system are summarized in **Error! Reference source not found..** These values indicate improvements that can be achieved through the consistent use of the digital twin for monitoring, analysis, and decision-making support in warehouse operations.

Table 2 Performance Outcomes and Operational Impact of the Proposed AWS-Based Digital Twin Solution

Category	Performance Outcome	Impact
Energy Efficiency	18-22% reduction in total warehouse energy consumption	Achieved by predictive maintenance minimizing equipment idle time and optimizing energy usage of forklifts, conveyors, and lighting systems.
Equipment Downtime	28-35% reduction in unplanned maintenance-related downtime	Enabled through real-time anomaly detection and predictive analytics using Amazon SageMaker.
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Sustainability Impact	8-10% water and resource usage reduction	As a result of improved machine scheduling and decreased idle heat production, the consumption of cooling and cleaning water reduced

These results suggest that the digital twin offers advantages that extend beyond mere observation. It enhances precision, agility, and resource efficiency, thereby directly aligning with the business goals outlined in previous sections.

5.5.2 Estimated Monthly Cost Breakdown

The estimated monthly operational expense for the proposed system was calculated using the AWS Pricing Calculator based on a scenario involving approximately 500 IoT sensors sending data to the cloud. Table 3 shows the cost contribution from each primary AWS service.

Table 3 Cost Estimation Breakdown

AWS Service	Usage Details	Estimated Monthly Cost (USD)
AWS IoT Core	500 devices sending data every 10 seconds (~129M messages/month)	\$215
Amazon Kinesis Data Streams	10 shards for storage and real-time data streaming	\$150
AWS Lambda	5M function invocations for data transformation and anomaly detection	\$25
Amazon DynamoDB	100 GB structured data + 5 million reads/writes	\$85
Amazon S3	500 GB data storage + 2 TB retrievals for analytics and visualization	\$55
AWS IoT SiteWise	100 assets, 250M data points/month	\$135
AWS IoT TwinMaker	1 workspace, 5 scenes, and 20 connected components	\$180
Amazon SageMaker	1 notebook instance + 50 training hours + hosted model endpoint	\$310
Amazon QuickSight	5 authors and 20 readers accessing dashboards	\$70
AWS CloudWatch	10 dashboards and 50 performance alarms	\$40
Amazon SNS	100,000 notifications per month for alerts	\$5

Total Estimated Monthly Cost: \$1,270/month

This projected expense represents consistent usage within standard warehouse operations. Due to architecture's dependence on managed and serverless services. This enables the system to stay cost-effective during times of reduced activity while still accommodating greater demands when warehouse throughput rises.

5.5.3 Cost Optimization Opportunities

The architecture of AWS enables the dynamic application of various cost-saving mechanisms to ensure efficiency as operations grow. The primary strategies consist of:

- **IoT Message Batching and Compression:** Lowers the cost of using AWS IoT Core by roughly fifteen percent through a reduction in the frequency of message transmissions.

- **S3 Intelligent Tiering:** Automatically transfers older or less often accessed data to more economical storage classes with savings of up to twenty percent on data storage.
- **On Demand SageMaker Endpoints:** Disabling inference endpoints when they are not in use can lead to a reduction in the cost of monthly machine learning hosting by approximately \$100-\$150.
- **AWS Budgets and Cost Explorer Integration:** This practice smooths the ongoing oversight of resource usage, automated notifications about costs, and anticipatory budgeting to ensure financial management.

Following the implementation of these optimizations, the anticipated monthly expenses are decreased from roughly one thousand one hundred dollars to an estimated \$850–\$950, thereby enhancing the financial efficiency of the system for medium-scale warehouse operations.

Scalability remains a central design principle of the aimed architecture:

- **Elastic Compute and Storage:** Amazon S3, Lambda, and DynamoDB doesn't require any manual provisioning as it automatically adjusts their capacity in response to data and demand volume.
- **Multi-Warehouse Integration:** AWS IoT TwinMaker and IoT Core can simultaneously be used to model and coordinate multiple warehouse locations.
- **Distributed Architecture:** The deployment across multiple availability zones guarantees resilience and ensures system uptime > 99.9%.
- **Serverless Framework:** This approach decreases maintenance efforts and aids swift growth in connected devices or processing capabilities as business requirements change.

Together, these characteristics guarantee that the digital twin system, as it evolves from a single warehouse configuration to a widespread logistics network solution, remains adaptable, sustainable, and economically viable.

6 Challenges and Future Improvements

The AWS-based digital twin architecture designed for the warehouse environment pointed out some implementation challenges. One of the major difficulties was the one related to the data quality of IoT sensors which is huge in quantity and placed at different locations. The entire system is dependent on the data to be both continuous and accurate in order to carry out the monitoring and analytics functions in real-time as well as predict the future state. A small mismatch in sensor calibration, network reliability, or data transmission can result in the loss of insights or predictions that are wrong. Device onboarding, sensor calibration, and data validation will have to be standardized so that they can be implemented, and the reliability ensured at the desired scale.

Coordinating the integration of existing warehouse management systems and legacy automated systems is a major challenge at the same time. A large number of warehouses use systems that have different data structures, interfaces, and synchronization cycles. A digital twin based on cloud

computing will require middleware or custom APIs that are able to translate and mix operational data in an uniform manner. The creation of secure, modular connectors would facilitate the adoption of the system without impacting on the regular workflow.

User adoption and operational training are constantly issues that have to be dealt with. A digital twin, in terms of its success, is not only related to the technical side of things, but also to onsite personnel's understanding and reaction to the insights produced. There may be changes in the workflow and collaboration between the analytical and operational teams in the coming of the new system. Easy-to-use dashboards, notifications that convey information, and the systematic training programs would all contribute to the development of trust and the prevention of technology being used to a lesser extent than intended.

Future enhancements can be done in the areas of automation, intelligence, and coverage as the major focus. For now, the architecture that is in place can only have its predictive maintenance and monitoring for one warehouse. It would be advantageous to have the whole logistics network shown on one integrated dashboard through managing multiple warehouses under a single system. The analytics layer can be improved by integrating additional operational and environmental data, which will allow Amazon SageMaker for deeper learning models and to support energy optimization , dynamic routing, and real-time decision recommendations. The control feedback loops incorporation would make a digital twin capable of automatically adjusting equipment or workflows per data insights, thus getting the system into the nearest self-optimizing warehouse environment.

7 Conclusion

This proposed digital twin solution based on AWS enhances the warehouse activities by providing uninterrupted data visibility, foresight, and decision-making support in real-time. The setup merges IoT sensing, real-time data processing, asset modeling, and visualization as a single cloud platform that enlightens and thus cuts downtime.

The assessment confirmed that the approach is still large-scale, dependable, and economical for mid-sized storage facilities. The AWS-based framework encompasses IoT detection, real-time data analysis, and digital twin visualization in a single, flexible platform that complies with the six Well-Architected pillars Performance Efficiency, Operational Excellence, Cost Optimization, Security, Reliability, and Sustainability and therefore protects, fortifies, simplifies, and takes care of the environment in system design.

Overall, the project shows that the digital twin technology can be successfully implemented with AWS managed services. The design laid out gives a strong architectural base which can expand to the networks of more than one warehouse and will accommodate future automation, advanced analytics, and smart, data-driven warehouse ecosystems.

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